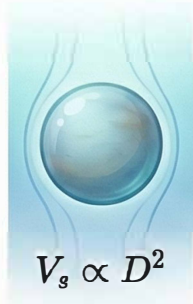


The Physics of Particle Settling Velocity



Stokes' Law for Fine Particles

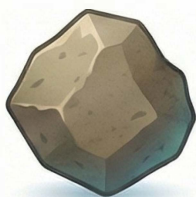
For small particles ($Re < 1$), settling velocity is proportional to the diameter squared.

Impact of Particle Shape

Angular or irregular natural sediments settle more slowly than perfectly smooth spheres.



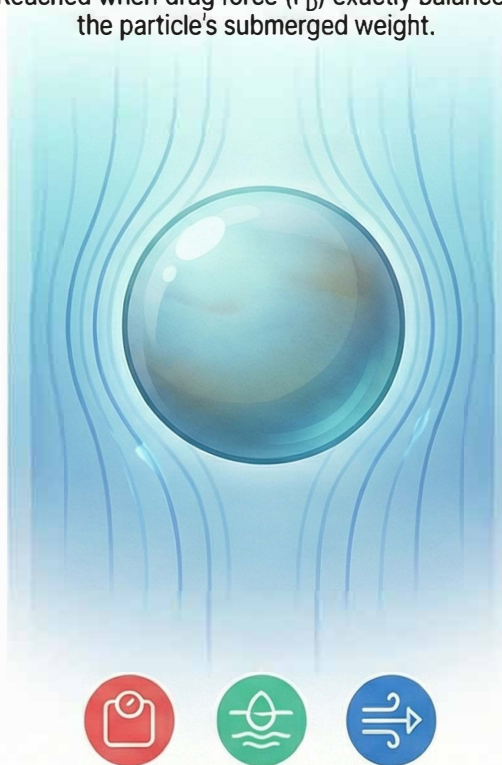
Faster Settling



Slower Settling

Terminal Velocity is a State of Equilibrium:

Reached when drag force (F_D) exactly balances the particle's submerged weight.

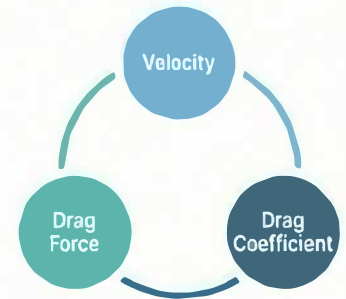


The Three Primary Vertical Forces

Settling is governed by the interaction of Particle Weight, Buoyant Force, and Drag.

The Iterative Calculation Paradox

Velocity depends on drag, but drag depends on velocity, requiring an iterative solving process.



Regime-Dependent Behavior & Flow Dynamics

The Reynolds Number determines if drag is dominated by fluid friction or pressure.

Hydrodynamic Regime	Reynolds Number (Re)	Dominant Physics
Stokes (Viscous)	$Re \leq 1$	Fully laminar flow; drag is dominated by viscous shear.
Transitional	$1 < Re < 10^3$	Flow separation begins, drag transitions from viscous to pressure-dominated.
Inertial	$Re \geq 10^3$	Pressure drag dominates, a turbulent wake forms behind the particle.